Paper 3.50 - Triality Surface Claim Contract

CQE-paper-03.50
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-03.50.md

Paper 3.50 - Triality Surface Claim Contract

Purpose

Paper 3.50 defines what counts as a valid claim about the Paper 3 registration surface. It preserves the useful local theorem while preventing stronger triality language from being silently counted as proved.

Admitted Paper 3 Claims

The following claims are admitted from Paper 3:

```
the eight LCR states have a bijective axis/sheet encoding
axis pairs are antipodal complement pairs
diag(L,C,R) preserves shell as trace
shell-2 states are trace-2 diagonal idempotents
Paper 2 correction coordinates are preserved as (2,0) and (3,1)
```

Claim Requirements

Any later paper using Paper 3 must state:

```
which LCR state or state set is being registered
which coordinate language is being used
whether the import is LCR, axis/sheet, diagonal trace, or idempotent
which receipt proves the imported row
whether any stronger D4/F4/J3(O) claim is proved separately
```

Linked Receipt

The minimum receipt link for Paper 3 is:

```
paper: CQE-paper-03
theorem: Local D4/J3 triality surface
receipt: production/formal-papers/CQE-paper-03/triality_surface_receipt.json
status: pass
```

The receipt is sufficient for finite registration. It is not sufficient by itself for full D4 triality, full F4 action, or full exceptional Jordan algebra dynamics.

Boundary Failures

The following are boundary failures:

```
using triality language without naming the finite registration actually proved
claiming Paper 3 proves full D4 triality
using off-diagonal octonionic claims without a separate receipt
changing the axis/sheet chart without rerunning the verifier
using Paper 2 correction rows without preserving their coordinates
```

Boundary failures become obligations for later group-action, repair, or causal-code papers.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt
predicted coordinate or scope status
revealed receipt
match/mismatch
next obligation if mismatch
```

The "does this prove full D4 triality?" prompt is required because the correct answer is negative. That negative result is part of the honesty layer.

Conclusion

Paper 3.50 lets later papers import the triality surface honestly. It keeps the finite object-registration theorem strong while making the unclosed exceptional-algebra extensions visible as separate obligations.